

Srinivas Sivakumar

AI / ML ENGINEER | DATA ENGINEER | MLOPS / DEVOPS
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AI/ML and Data Engineer building production-focused AI agents, multimodal/computer-vision systems, predictive models, and cloud/data infrastructure across manufacturing, networking, conservation, and fashion.

CORE SKILLS

AI/ML: PyTorch, TensorFlow, Hugging Face, LLMs, RAG, FAISS, CLIP, Computer Vision, Diffusion, MLflow | **Backend/Data:** FastAPI, PostgreSQL, Spark, Snowflake, Airflow, Dagster

Cloud/DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD, Railway, OpenStack, Linux | **Languages:** Python, SQL, TypeScript, Bash

PROFESSIONAL EXPERIENCE

AI Software Engineer - Mexo | *Very Early-Stage Startup* Jun 2026 - Aug 2026

- Early engineering hire on a two-person team; built a manufacturing AI agent with TypeScript, Flue, OpenRouter and PostgreSQL to analyze production progress, capacity, downtime and dependencies, surfacing delivery risks with evidence-backed explanations.
- Built Terraform-managed AWS S3 backup infrastructure and migrated database backups and historical data to centralized object storage.

R&D Engineer (AI) - Zybasis Dec 2025 - May 2026

- Developed an AI bandwidth-prediction and network anomaly-detection workflow using Python and CheckMK telemetry, applying client/time-specific baselines, tolerance-band analysis and multi-metric correlation across bandwidth, utilization, latency, IOPS and errors/drops.
- Built a deterministic + LLM-assisted RCA pipeline using runbook pattern matching, fault-domain classification and Claude API fallback to identify client-side/internal incidents and generate evidence-backed troubleshooting and escalation actions.

Data Engineer - Wildlife Conservation Trust Sep 2024 - Aug 2025

- Deployed and maintained RStudio Shiny servers with CI/CD automation using Jenkins, Airflow, and Dagster, enabling scientists to continuously access and visualize research data.
- Built AWS-based pipelines using S3, Lambda, Airflow, and Snowflake for real-time sensor data processing and delivery of Chambal River monitoring data.
- Developed image recognition systems with automated training and reporting pipelines to identify individual crocodiles during wildlife surveys.
- Created a GPS-free 3D tracking system using Python, AWS EC2, and a hybrid OpenStack deployment as a prototype for tracking underwater Gangetic dolphins.

Data Engineering Intern - NCBS (TIFR) Jun 2023 - Aug 2024

- Deployed Ceph using Docker with automated backup pipelines integrated into OpenStack.
- Built PySpark ETL pipelines using Databricks, Airflow, Snowflake, and Dagster.
- Deployed GitLab on OpenStack with automated backup and access-control systems.

PROJECTS & PAID CLIENT ENGAGEMENT

Multimodal AI Fashion Assistant - MyMirro | *Paid Freelance Client Engagement* Sep 2025 - Nov 2025

- Built an end-to-end assistant that turns wardrobe images into a digital wardrobe and generates outfit, styling and shopping recommendations using Gemini, FashionCLIP/CLIP, FAISS, SegFormer and SAM.
- Deployed with FastAPI, MLflow, Docker, Terraform and Railway; built Python/TypeScript scraping pipelines aggregating 1,000+ products per run from 10+ e-commerce sites.

X-Pent Architecture - Independent Contribution 2026

- Designed a provider-neutral Docker Compose setup that lets X-Pent run locally and move across AWS, Azure, GCP or VPS environments with minimal changes, using PostgreSQL + pgvector, Redis, MinIO/S3-compatible storage, a RAG service skeleton and policy-based AI routing.

Microscopy Nuclei Detection - Computer Vision Project

- Built a Python/scikit-image pipeline using Gaussian smoothing, Otsu thresholding, morphology and region-property filters to isolate and outline nuclei in microscopy TIF images.

Pulmonary Fibrosis Progression Prediction - ML Project

- Built a TensorFlow/Keras model combining chest CT DICOM images with age, sex and smoking metadata through a ResNet-style CNN to predict the rate of FVC decline.

EDUCATION & CREDENTIALS

MCA - Storage & Cloud Computing, Jain University (2022-2024) | **BCA - Computer Applications**, Indira College, Pune University (2019-2022)

Cloud Big Data Engineering - TrendyTech Insights (Jul 2026) | **Linux Administration - Besant Technologies** | **Claude Certification - Claude** | **Design Patent:** light-based automatic aquarium fish feeder